

ZHENGTAO GAO

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Education

Università della Svizzera italiana (USI, Lugano, Switzerland)
Master of Science in Artificial Intelligence

Sept 2024 – Present
GPA: 9.5/10

Nankai University (Tianjin, China)
Bachelor of Science (Major: Chemical Biology)

Sept 2019 – Jun 2023
GPA: 3.7/4.0 (87.98/100)

Publications

- [1] H. Jing, **Z. Gao**, S. Xu, T. Shen, Z. Peng, S. He, T. You, S. Ye, W. Lin, S. Sun. Accurate prediction of antibody function and structure using bio-inspired antibody language model. In: *Briefings in Bioinformatics* 25.4 (2024). DOI: [10.1093/bib/bbae245](https://doi.org/10.1093/bib/bbae245).

Research Experience

Cross-User Generalization and Personalization for Multimodal Smart-Glass Data

Jul 2025 – Mar 2026

↪ Research Assistant, Advisor: Dr. Martin Gjoreski

People-Centered Computing Lab, USI

- Identified inter-subject domain shift as a key challenge in multimodal smart-glass sensor data, which leads to significant performance degradation when generalizing to unseen users.
- Pretrained a ViT model using MAE on multimodal sensor data augmented with Euler angle features, and leveraged user identity as a zero-cost label to investigate its effect on cross-user generalizability.
- Fine-tuning the pretrained model with only 5% of labeled data achieves comparable or superior performance to a fully supervised baseline trained on 100% of labeled data, across two downstream tasks.

Contrastive Learning based Molecular Clustering [\[GitHub Link\]](#)

Jan 2023 – May 2023

↪ Graduation Thesis, Advisor: Prof. Wensheng Cai

Laboratory of Chemoinformatics, Nankai University

- Adapted the SwAV contrastive framework to molecular dynamics, designing a GNN to extract end-to-end representations directly from raw 3D coordinates by enforcing translational invariance via zero-centering and empirical rotational invariance through $SO(3)$ data augmentation.
- Benchmarked on alanine dipeptide trajectories, demonstrating that the learned representations effectively capture the distinct metastable states present in the classical free-energy landscape.

Bio-Inspired Antibody Language Model (BALM)

Jul 2022 – Jan 2023

↪ Research Assistant, Advisor: Prof. Siqi Sun

Institute of Intelligent Complex Systems, Fudan University

- Motivated by the limitation of general protein language models like ESM-2 in capturing antibody-specific properties due to their unique evolutionary patterns and high flexibility in antigen-binding regions.
- Designed the BALM architecture to incorporate biological inductive biases by introducing region-aware positional encodings and conservation-based masking strategies.
- Preprocessed 1B+ sequences and conducted distributed pre-training on 336M non-redundant samples, achieving SOTA performance across functional properties and structure prediction tasks.

Awards

Foundation for the Lugano Faculties Scholarship, USI Awarded to the best student in Master's Program. 2025

Foundation for the Lugano Faculties Scholarship, USI Selected as 1 of 60 recipients via application. 2024

Academic Excellence Scholarship, Nankai University 2019 – 2020, 2020 – 2021

Selected Projects

Reproduction of Stochastic Concept Bottleneck Models *Advanced Topics in Machine Learning* Spring 2026

- Reproduced CBM, CEM, Autoregressive CBM, and SCBM on synthetic, CUB, and CIFAR-10 benchmarks, analyzing concept prediction, target accuracy, calibration, and intervention behavior.
- Refactored the original research codebase into modular training and intervention pipelines, and resolved key performance bottlenecks through vectorized concept modules, batched Monte Carlo sampling, optimized data loading, and GPU-accelerated SCBM intervention solvers.
- Reduced estimated full experimental runtime from 1,200 hours to around 60 hours while preserving experimental semantics, making future research on CBMs much more accessible and efficient.

Interactive Cooking Agent *Course Project, Natural Language Processing*

Autumn 2025

- Built a RAG pipeline indexing 2.2M recipes using FAISS (IVF quantization) and sentence transformers, powering semantic retrieval for a locally deployed Qwen3-8B model via Ollama.

- Designed a multi-node LangGraph agent with intent classification, context-aware memory summarization, and step-by-step interactive guidance, enabling real-time conversational cooking assistance.
- Conducted a qualitative LLM-as-judge evaluation using Gemini 3 Pro, confirming the designed agent scored 5/5 in interactive experience versus 1/5 for a naive retrieval-only baseline.

Reproduction of g-GRFs *Reproducibility Challenge, Graph Deep Learning* [\[GitHub Link\]](#) **Autumn 2025**

- Reproduced all five experiments from *General Graph Random Features* (ICLR 2024), extending the original codebase which only provided a partial implementation for Experiment 1.
- Achieved > 10× speedup for Experiment 1 through Numba JIT compilation and cache optimization.

Photon Mapping Renderer (C++) *Rendering Competition, Computer Graphics* [\[GitHub Link\]](#) **Autumn 2025**

- Implemented photon mapping with KD-tree acceleration for global illumination in a C++ ray tracer.
- Invited to present at the USI Autumn Semester Project Presentations.

Monte Carlo Localization Simulator (Python) *Course Project, Robotics* [\[GitHub Link\]](#) **Spring 2025**

- Implemented a configurable Monte Carlo Localization simulator in Python, integrating differential drive kinematics, ray-casting distance sensing, and Gaussian likelihood weighting.
- Accelerated ray intersection computation by 10× and reduced overall rendering time by 50% by replacing Shapely API calls with custom vectorized NumPy cross-product logic.
- Improved tracking robustness and prevented particle depletion during sharp turns by designing a hybrid Low-Variance resampling strategy paired with exponential moving average weight smoothing.

Technical Skills

Programming Languages	Python, C++, MATLAB
ML / DL	Transformers, Representation Learning, Self-supervised learning, GNN
Python Libraries	PyTorch, NumPy, pandas, Matplotlib, LangChain and LangGraph
Systems & Tools	Linux, Git, L ^A T _E X
Languages	English (fluent), Italian (basic), Chinese (native)